Detailed Technical Specifications with Numerical Targets and Tolerances

Enhanced Plant Communication Sensor Network - Manship Forest Installation

1. SYSTEM ARCHITECTURE SPECIFICATIONS

1.1 Network Topology Requirements

- **Network Type**: LoRa mesh network with star topology expandable to mesh
- **Coverage Area**: 15 acres (60,703 m²) minimum, 25 acres (101,171 m²) maximum
- **Communication Range**: 2.0 km minimum in forested terrain ±200m
- **Node Count**: 7 nodes Phase 1, expandable to 25 nodes without infrastructure changes
- **Gateway Count**: 1 primary gateway, expandable to 3 gateways for redundancy
- **Network Capacity**: 1,000 devices maximum per gateway

1.2 Data Transmission Specifications

- **Data Rate**: 1 kbps average, 10 kbps burst capacity
- **Packet Size**: 51 bytes maximum per LoRa packet (LoRaWAN SF7-SF12)
- **Transmission Interval**: 3600 seconds (1 hour) standard, 60 seconds emergency mode
- **Packet Loss Tolerance**: <1.0% under normal conditions, <5.0% during severe weather
- **Latency Requirements**: <500ms node-to-cloud for emergency alerts, <100ms for artistic interfaces
- **Network Uptime**: >95.0% availability over 12-month period

2. PROCESSING HARDWARE SPECIFICATIONS

2.1 Primary Microcontroller (ESP32-C3-WROOM-02)

- **Processor**: 32-bit RISC-V single-core @ 160MHz ±5%
- **Memory**: 400KB SRAM, 4MB Flash storage
- **Power Consumption**:
  - Active mode: 50-80mA @ 3.3V
  - Deep sleep: <10µA @ 3.3V
  - Target average: <50mA including sensor operation

- **Operating Voltage**: 3.0-3.6V DC
- **Operating Temperature**: -40°C to +85°C (industrial grade)
- **WiFi Specification**: 802.11 b/g/n, 2.4GHz band
- **Bluetooth**: Bluetooth 5.0 LE
- **GPIO Pins**: 22 available (14 used for sensors)

2.2 LoRa Communication Module (SX1276)

- **Frequency**: 915MHz ±13MHz (ISM band, North America)
- **Output Power**: +20dBm maximum, configurable 2-20dBm
- **Sensitivity**: -137dBm (SF12, 125kHz BW)
- **Link Budget**: 157dB maximum
- **Modulation**: LoRa CSS (Chirp Spread Spectrum)
- **Spreading Factor**: SF7-SF12 (configurable for range vs. data rate optimization)
- **Bandwidth**: 125kHz, 250kHz, 500kHz selectable
- **Current Consumption**: 120mA TX @ +20dBm, 10.3mA RX mode, 0.2µA sleep

2.3 Analog-to-Digital Converter (ADS1115)

- **Resolution**: 16-bit ADC
- **Sampling Rate**: 860 samples per second maximum
- **Input Channels**: 4 differential or 8 single-ended
- **Programmable Gain**: ±6.144V, ±4.096V, ±2.048V, ±1.024V, ±0.512V, ±0.256V
- **Interface**: I2C (400kHz max clock)
- **Accuracy**: ±0.5 LSB integral non-linearity
- **Power Supply**: 2.0V to 5.5V

3. ENVIRONMENTAL SENSOR SPECIFICATIONS

3.1 Soil Moisture Sensor (Capacitive Type)

- **Measurement Range**: 0-100% volumetric water content (VWC)
- **Accuracy**: ±2.0% VWC after calibration
- **Resolution**: 0.1% VWC
- **Response Time**: <10 seconds for 95% reading
- **Operating Temperature**: -30°C to +70°C
- **Probe Length**: 100mm ±5mm insertion depth
- **Output**: 0-3.0V analog (linear response)

- **Power Consumption**: <3mA during measurement
- **Calibration**: 3-point calibration (air, distilled water, soil standard)
- **Long-term Stability**: ±1% drift over 12 months

3.2 Temperature and Humidity Sensor (DHT22)

- **Temperature Range**: -40°C to +80°C
- **Temperature Accuracy**: ±0.5°C (typical)
- **Temperature Resolution**: 0.1°C
- **Humidity Range**: 0-100% RH
- **Humidity Accuracy**: ±2% RH (typical), ±5% RH (maximum)
- **Humidity Resolution**: 0.1% RH
- **Sampling Rate**: 0.5Hz (one measurement every 2 seconds)
- **Response Time**: <2 seconds (τ63%)
- **Operating Voltage**: 3.3-6.0V DC
- **Current Consumption**: 1-1.5mA during measurement, 40-50μA standby
- **Digital Interface**: Single-wire digital communication

3.3 Light Intensity Sensor (BH1750FVI)

- **Measurement Range**: 1-65,535 lux
- **Spectral Response**: Peak sensitivity at 560nm (photopic response)
- **Accuracy**: ±20% typical (relative to standard illuminant A)
- **Resolution**: 1 lux (high resolution mode), 4 lux (low resolution mode)
- **Measurement Time**: 120ms (high resolution), 16ms (low resolution)
- **Operating Temperature**: -40°C to +85°C
- **Interface**: I2C (400kHz maximum clock)
- **Power Consumption**: 120μA active, 0.01μA power down
- **Calibration**: Factory calibrated, field verification with lux meter

3.4 Soil pH Sensor (Ion-Selective Electrode)

- **Measurement Range**: pH 4.0 to 10.0
- **Accuracy**: ±0.1 pH units after 2-point calibration
- **Resolution**: 0.01 pH units
- **Response Time**: <30 seconds for 95% reading
- **Temperature Compensation**: Manual or automatic (0-100°C)
- **Electrode Type**: Glass membrane pH electrode with reference

- **Operating Temperature**: 0°C to 60°C (aqueous solutions)
- **Calibration**: 2-point calibration using pH 4.01 and pH 7.00 buffers
- **Electrode Life**: >2 years with proper storage and maintenance
- **Output**: 0-3.3V (via pH measurement IC)

3.5 Atmospheric Pressure Sensor (BMP280)

- **Pressure Range**: 300-1100 hPa (9000m to -500m altitude)
- **Pressure Accuracy**: ±1 hPa absolute, ±0.12 hPa relative
- **Pressure Resolution**: 0.01 hPa (0.08 Pa)
- **Temperature Accuracy**: ±1.0°C
- **Temperature Resolution**: 0.01°C
- **Interface**: I2C and SPI
- **Power Consumption**: 2.7µA @ 1Hz sampling
- **Operating Temperature**: -40°C to +85°C
- **Response Time**: <1 second

4. PLANT COMMUNICATION SENSOR SPECIFICATIONS

4.1 Bioelectrical Signal Detection System

- **Electrode Type**: Gold-plated stainless steel needle electrodes
- **Electrode Spacing**: 10mm ±1mm between measurement points
- **Input Impedance**: >10MΩ differential input
- **Voltage Range**: ±100mV full scale
- **Amplification**: 2-stage instrumentation amplifier, 1000x total gain ±10%
- **Frequency Response**: 0.01Hz to 50Hz (-3dB points)
- **Noise Floor**: <5µV RMS (0.01-50Hz bandwidth)
- **Common Mode Rejection**: >80dB @ 50Hz
- **Sampling Rate**: 100Hz ±1% (configurable 10Hz-1kHz)
- **Resolution**: 12-bit ADC (24µV per LSB after amplification)
- **Power Supply**: ±5V for analog front-end, isolated from digital

4.2 Volatile Organic Compound (VOC) Detection Array

4.2.1 General VOC Sensor (TGS2600)

- **Target Gases**: General VOCs, alcohols, organic solvents
- **Detection Range**: 1-30 ppm (ethanol equivalent)
- **Sensitivity**: 0.3-0.8 (Rs/Ro in 10ppm ethanol)
- **Response Time**: <30 seconds (τ90%)
- **Operating Temperature**: -10°C to +50°C
- **Operating Humidity**: 65% ±5% RH (optimal)
- **Power Consumption**: 42mW ±10% (heater power)
- **Output**: Analog resistance change (measured via voltage divider)

4.2.2 Methane Sensor (TGS2611-C00)

- **Target Gas**: Methane (CH4)
- **Detection Range**: 500-10,000 ppm
- **Sensitivity**: 1.0-3.0 (Rs/Ro in 1000ppm methane)
- **Response Time**: <30 seconds (τ90%)
- **Cross-Sensitivity**: <1% to CO, <2% to alcohol
- **Operating Temperature**: -10°C to +50°C
- **Power Consumption**: 56mW ±10%

4.2.3 Hydrogen Sulfide Sensor (TGS2603)

- **Target Gas**: Hydrogen sulfide (H2S)
- **Detection Range**: 0.1-10 ppm
- **Sensitivity**: 0.6-0.9 (Rs/Ro in 1ppm H2S)
- **Response Time**: <30 seconds
- **Operating Temperature**: -10°C to +50°C
- **Power Consumption**: 42mW ±10%

4.2.4 Additional VOC Sensors (TGS2442, TGS2444, TGS2620, TGS2601, TGS2606)

- **Carbon Monoxide (TGS2442)**: 1-300 ppm range, 165mW power
- **Ammonia (TGS2444)**: 1-300 ppm range, 165mW power
- **Alcohol/Ethanol (TGS2620)**: 50-5000 ppm range, 42mW power
- **Formaldehyde (TGS2601)**: 1-100 ppm range, 42mW power
- **Ozone (TGS2606)**: 0.05-10 ppm range, 42mW power

4.3 Ultrasonic Acoustic Monitoring

- **Microphone Type**: MEMS omnidirectional microphone (SPU0410LR5H-QB)
- **Frequency Response**: 20Hz-100kHz ±3dB
- **Sensitivity**: -38dBV/Pa ±3dB
- **Signal-to-Noise Ratio**: >61dB(A)
- **Maximum SPL**: 120dB SPL @ 1% THD
- **Power Supply**: 1.5-3.6V
- **Current Consumption**: 230µA typical
- **Sampling Rate**: 200kHz (configurable down to 8kHz)
- **Dynamic Range**: >100dB
- **Acoustic Isolation**: Custom housing with foam windscreen

5. ENHANCED SENSING SPECIFICATIONS

5.1 Chlorophyll Fluorescence System (Fluorometer)

5.1.1 Excitation LED Specification

- **Wavelength**: 470nm ±10nm (blue LED)
- **Optical Power**: 1W maximum, software controlled
- **Pulse Duration**: 0.1-10 seconds programmable
- **Rise Time**: <1ms to 90% intensity
- **Spectral Half-Width**: 25nm ±5nm
- **Operating Life**: >50,000 hours @ 25°C

5.1.2 Detection Photodiode Specification

- **Spectral Response**: 400-1000nm
- **Peak Sensitivity**: 700-800nm region
- **Active Area**: 5.8mm² minimum
- **Dark Current**: <2nA @ 25°C
- **Response Time**: <1µs
- **Operating Temperature**: -40°C to +100°C

5.1.3 Optical Filter System

- **Long-pass Filter**: 700nm cut-on ±10nm
- **Transmission**: >95% @ 720-800nm
- **Blocking**: OD4 (0.01% transmission) @ 400-680nm
- **Temperature Stability**: ±0.1nm/°C

5.1.4 Fluorometer Performance Specifications

- **Measurement Parameters**: Fo, Fm, Fv/Fm, ΦPSII, NPQ
- **Fv/Fm Range**: 0.1-0.9 (typical healthy plants: 0.75-0.85)
- **Measurement Accuracy**: ±0.02 Fv/Fm units
- **Response Time**: <30ms for complete measurement
- **Signal-to-Noise Ratio**: >50:1 for healthy leaf samples
- **Sampling Frequency**: 0.1-10Hz configurable

5.2 Electronic Nose Data Processing

- **Sensor Array Size**: 8 MEMS gas sensors
- **Sampling Rate**: 1Hz per sensor (staggered measurements)
- **Pattern Recognition**: Support Vector Machine (SVM) classification
- **Training Dataset**: Minimum 100 samples per plant stress condition
- **Classification Accuracy**: >85% for trained stress patterns
- **Response Matrix**: 8×8 sensor response correlation analysis
- **Drift Compensation**: Weekly baseline recalibration
- **Processing Platform**: Edge Impulse ML framework on ESP32

6. POWER SYSTEM SPECIFICATIONS

6.1 Solar Panel Specifications (Per Node)

- **Rated Power**: 50W ±5% under Standard Test Conditions (STC)
- **STC Conditions**: 1000W/m², 25°C cell temperature, AM1.5 spectrum
- **Voltage at Maximum Power (Vmp)**: 17.8V ±5%
- **Current at Maximum Power (Imp)**: 2.81A ±5%
- **Open Circuit Voltage (Voc)**: 22.1V ±5%
- **Short Circuit Current (Isc)**: 3.00A ±5%
- **Cell Type**: Monocrystalline silicon
- **Efficiency**: >18% module efficiency

- **Temperature Coefficient**: -0.41%/°C (power), -0.33%/°C (Voc)
- **Operating Temperature**: -40°C to +85°C
- **Mechanical Load**: 2400 Pa wind load, 5400 Pa snow load

6.2 Battery System Specifications

- **Battery Type**: Lithium-ion 18650 cells (2 cells per node)
- **Nominal Capacity**: 3500mAh per cell (7Ah total per node)
- **Nominal Voltage**: 3.7V per cell (series/parallel configuration)
- **Operating Voltage**: 2.5V to 4.2V per cell
- **Discharge Rate**: 0.5C maximum continuous
- **Cycle Life**: >2000 cycles @ 80% depth of discharge
- **Self-discharge**: <3% per month @ 20°C
- **Operating Temperature**: -20°C to +60°C (charge: 0°C to +45°C)
- **Protection**: Built-in PCB with over-voltage, under-voltage, over-current protection

6.3 Charge Controller Specifications

- **Type**: PWM (Pulse Width Modulation) charge controller
- **Maximum Input**: 75V, 10A
- **Battery Voltage**: 12V system (configurable for 3.7V Li-ion)
- **Charge Efficiency**: >95% @ rated conditions
- **Load Disconnect**: 10.5V (reconnect 12.6V) for 12V system
- **Temperature Compensation**: -30mV/°C for lead-acid (disabled for Li-ion)
- **Protection Features**: Reverse polarity, over-load, short-circuit protection
- **Quiescent Current**: <4mA

6.4 Power Budget Analysis (Per Node)

- **Total Average Power Consumption**: <2.5W (24-hour average)
- **Component Breakdown**:
  - ESP32-C3 (active 10%, sleep 90%): 8mA average
  - LoRa module (TX 5 min/hour): 3mA average
  - Environmental sensors (duty cycled): 15mA average
  - Plant communication sensors: 20mA average
  - VOC sensor array (heated): 25mA average
  - Fluorometer (pulsed operation): 5mA average
- **Peak Power**: <5W during simultaneous sensor operation and LoRa transmission

- **Autonomy Target**: >60 days without solar input (worst-case scenario)
- **Solar Energy Budget**: 200Wh/day average production, 60Wh/day consumption

7. ENVIRONMENTAL PROTECTION SPECIFICATIONS

7.1 Enclosure Requirements

- **Ingress Protection**: IP67 rating minimum (dust-tight, immersion to 1m)
- **Material**: UV-stabilized polycarbonate or ABS plastic
- **Operating Temperature**: -30°C to +70°C
- **Thermal Expansion**: <0.1mm dimensional change over temperature range
- **UV Resistance**: >10 years outdoor exposure (ASTM G154)
- **Impact Resistance**: IK08 rating (5 Joule impact energy)
- **Dimensions**: 200mm × 150mm × 100mm maximum per node enclosure
- **Mounting**: Tree-friendly clamp system, no drilling required
- **Cable Entries**: IP68-rated cable glands for all external connections

7.2 Environmental Operating Conditions

- **Ambient Temperature Range**: -20°C to +60°C operational
- **Storage Temperature**: -40°C to +85°C non-operational
- **Humidity Range**: 5% to 95% RH non-condensing
- **Condensation**: Desiccant sachets and breather vents for pressure equalization
- **Wind Loading**: Survive 160 km/h (100 mph) wind gusts
- **Precipitation**: Continuous operation during rainfall up to 200mm/day
- **Snow Loading**: Support up to 50kg distributed load on solar panels
- **Vibration**: MIL-STD-810G Method 514.6 (transportation and operational)

7.3 Electromagnetic Compatibility (EMC)

- **EMI Emissions**: FCC Part 15.109, Class B limits
- **EMI Susceptibility**: EN 61000-4-3 (radiated immunity, 10V/m)
- **ESD Protection**: EN 61000-4-2 (±15kV air discharge, ±8kV contact)
- **Power Line Transients**: EN 61000-4-5 (±2kV line-to-line, ±4kV line-to-ground)
- **RF Immunity**: 915MHz ±13MHz ISM band, co-location with multiple LoRa devices

8. DATA COLLECTION AND PROCESSING SPECIFICATIONS

8.1 Sampling Rates and Data Resolution

- **Environmental Sensors**: 0.1Hz (6-minute intervals) ±10%
- **Bioelectrical Signals**: 100Hz ±1% (configurable 10Hz-1kHz)
- **VOC Sensor Array**: 1Hz per sensor (8-second full array scan)
- **Fluorometer Measurements**: 0.017Hz (1 measurement per minute)
- **Ultrasonic Recording**: 200kHz sample rate, 10-second bursts every 10 minutes
- **Data Resolution**: 16-bit ADC for all analog measurements
- **Timestamp Accuracy**: ±1 second (GPS or NTP synchronized)

8.2 Local Data Storage

- **Storage Capacity**: 4MB flash memory per node
- **Data Retention**: 30 days minimum at full sampling rates
- **Compression**: DEFLATE algorithm for time-series data (50-70% compression ratio)
- **Data Format**: JSON for configuration, binary for sensor data
- **Backup Strategy**: Redundant storage on gateway and cloud systems
- **Data Integrity**: CRC32 checksums for all stored data packets

8.3 Real-time Processing Requirements

- **Edge Processing**: Moving average filters, threshold detection, anomaly flagging
- **Latency Requirements**: <100ms for artistic interface data paths
- **Processing Capacity**: 32-bit RISC-V @ 160MHz with dedicated DSP functions
- **Machine Learning**: TensorFlow Lite models for pattern recognition
- **Memory Usage**: <256KB RAM for real-time processing buffers
- **Interrupt Handling**: <10µs response time for critical sensor events

9. ARTISTIC INTERFACE SPECIFICATIONS

9.1 Max/MSP Integration Requirements

- **Protocol**: Open Sound Control (OSC) over UDP
- **Data Rate**: 30 messages per second maximum per node
- **Latency Target**: <50ms from sensor to audio synthesis
- **Data Format**: 32-bit floating point sensor values with metadata
- **Network Requirements**: Standard Ethernet or WiFi connection

- **Compatibility**: Max/MSP 8.0 or newer versions
- **API Documentation**: Complete OSC namespace and message format specification

9.2 TouchDesigner Integration Requirements

- **Protocol**: WebSocket with JSON message formatting
- **Frame Rate**: 30 FPS update rate for smooth visual feedback
- **Data Synchronization**: Multiple node coordination with master clock
- **Video Output**: Support for 1080p @ 60Hz minimum
- **Network Bandwidth**: <1 Mbps per node for visual data streams
- **Compatibility**: TouchDesigner 2022.30000 or newer

9.3 Mobile Application Requirements

- **Platform Support**: iOS 14+ and Android 8+ (API level 26+)
- **Framework**: React Native for cross-platform development
- **Real-time Updates**: WebSocket connection with <2-second update intervals
- **Offline Capability**: 24-hour data caching for areas with poor cellular coverage
- **User Interface**: Responsive design supporting 4.7" to 12.9" screen sizes
- **Authentication**: OAuth 2.0 with role-based access control

10. PERFORMANCE VALIDATION AND TESTING SPECIFICATIONS

10.1 Calibration Requirements

- **Environmental Sensors**: NIST-traceable standards for temperature, humidity, pressure
- **pH Sensors**: Certified buffer solutions (pH 4.01 ±0.01, pH 7.00 ±0.01)
- **Light Sensors**: Calibrated light sources with known lux output ±5%
- **VOC Sensors**: Certified gas standards with known concentrations ±2%
- **Calibration Frequency**: Annual recalibration for critical measurements
- **Field Calibration**: Monthly verification using portable reference instruments
- **Documentation**: Complete calibration certificates with traceability chain

10.2 Performance Testing Protocols

- **Environmental Chamber Testing**: -20°C to +60°C, 10% to 95% RH cycling
- **Vibration Testing**: MIL-STD-810G Method 514.6 Category 4 (general vibration)
- **EMC Testing**: FCC Part 15 Class B emissions and immunity testing
- **Communication Range**: Field testing with spectrum analyzer and RSSI mapping

- **Power Consumption**: 168-hour continuous monitoring with precision current measurement
- **Data Integrity**: Statistical analysis of 30-day continuous data collection
- **Network Stress Testing**: Simultaneous operation of all 7 nodes with packet loss monitoring

10.3 Acceptance Criteria

- **Sensor Accuracy**: All measurements within specified tolerance after calibration
- **Network Reliability**: >95% packet delivery over 30-day test period
- **Power Efficiency**: >60-day autonomous operation during no-solar conditions
- **Environmental Survival**: No degradation after 168-hour environmental cycling
- **Data Quality**: <0.1% data corruption rate in storage and transmission
- **Real-time Performance**: <100ms latency for 95% of artistic interface transactions
- **System Uptime**: >99% availability during 30-day acceptance test period

11. MAINTENANCE AND LIFECYCLE SPECIFICATIONS

11.1 Preventive Maintenance Schedule

- **Daily**: Automated system health checks and alert generation
- **Weekly**: Remote data quality assessment and battery voltage monitoring
- **Monthly**: Visual inspection of solar panels and enclosure integrity
- **Quarterly**: Physical inspection of all connections and sensor cleaning
- **Annual**: Complete calibration verification and component replacement assessment
- **Component Replacement**:
  - Batteries: 3-5 years (cycle life dependent)
  - Sensors: 2-5 years (environmental exposure dependent)
  - Electronics: 10+ years (proper maintenance assumed)

11.2 Diagnostic and Monitoring Capabilities

- **Self-Diagnostics**: Automated sensor functionality testing and fault reporting
- **Remote Monitoring**: Real-time system status via cloud dashboard
- **Predictive Maintenance**: Battery aging algorithms and performance trend analysis
- **Alert System**: Immediate notification for critical failures or performance degradation
- **Logging**: Comprehensive system logs with 6-month retention
- **Firmware Updates**: Over-the-air capability for bug fixes and feature enhancements

11.3 End-of-Life and Sustainability

- **Design Life**: 10 years minimum operational lifetime
- **Component Recyclability**: >85% materials recyclable (batteries, metals, plastics)
- **Data Preservation**: Complete system backup before decommissioning
- **Environmental Impact**: Minimal ecological footprint during removal
- **Technology Transfer**: Complete documentation package for system replication
- **Legacy Support**: 5-year post-deployment technical support commitment

This comprehensive technical specification document provides the numerical targets, tolerances, and detailed requirements necessary for professional development of the Enhanced Plant Communication Sensor Network. All specifications are based on commercially available components and proven technologies to ensure technical feasibility while maintaining research-grade performance standards.